61P - Neoadjuvant ipilimumab, nivolumab, and high-dose vitamin C in pMMR/MSS colon cancer: results of the ALFEO pilot study

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BACKGROUND

Immune checkpoint inhibitors (ICI) are inactive in proficient mismatch-repair (pMMR)/microsatellite-stable (MSS) colon cancer (CC), ~85% of cases. Preclinically, high-dose IV vitamin C (HDVC) increases CD8+ T-cell infiltration and synergizes with ICI in pMMR CC. In NICHE, nivolumab + ipilimumab gave ~20% major pathological response (MPR) in pMMR CC.[1] We tested whether adding HDVC restores ICI efficacy and dissected the immune correlates of response.

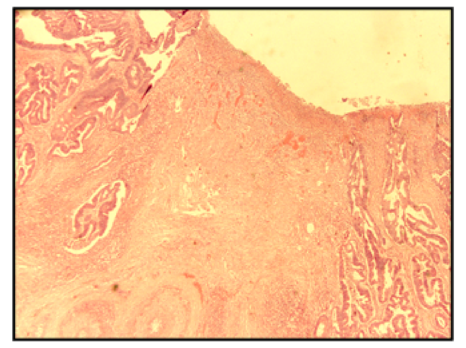
METHODS

ALFEO (EU CT 2022-502524-41) is an open-label, single-arm, multicentre **Bayesian (BOP2)** pilot trial. Eligible: pMMR/MSS CC, stage II–III or resectable liver-limited stage IV, candidates for R0 resection. Treatment: **ipilimumab 1 mg/kg d2 + nivolumab 3 mg/kg d2,16 + HDVC 70 g/m² d1–3, d15–17**, then surgery within 4–6 weeks. **Primary endpoint:** MPR (≤10% viable tumour, Mandard). Prior Beta(6.9, 23.1); H0=0.23, H1=0.75; futility stop at 12 pts if ≤2 MPR. Integrated translational platform: single-cell RNA-seq + TCR-seq (paired pre-ICI biopsy, post-ICI resection, PBMC), bulk peripheral TCR-β (Omniscope), and AI-driven computational pathology for tertiary lymphoid structures (TLS).

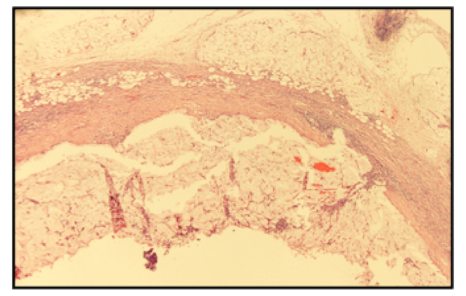
CLINICAL RESULTS

- **12 patients** were enrolled (Jun 2023–Jun 2025; median age 58; 25% stage IV). **11 underwent R0 resections**; 1 progressed pre-surgery (lung metastases).
- **MPR 2/12 (16.7%; 95% CI 2.1–48.4%).** Posterior **P(p_eff>0.23)=36.2% < 43%** decision threshold → the trial regularly **stopped for futility** at interim.
- Pathologic regression according to Mandard score was: G1 2 (18%), G4 7 (64%), G5 2 (18%); median residual viable tumour **80%** (IQR 33–90%).
- **Safety was manageable:** grade ≥3 AE 42%; 2 grade-4 SAE (peri-surgical ileal perforation; peri-operative hypoxia) resolved, **not treatment-related**; no grade 5. Compliance ICI 100%, HDVC 92%. HDVC PK confirmed therapeutic plasma levels with no vitamin-C–specific antitumour signal.

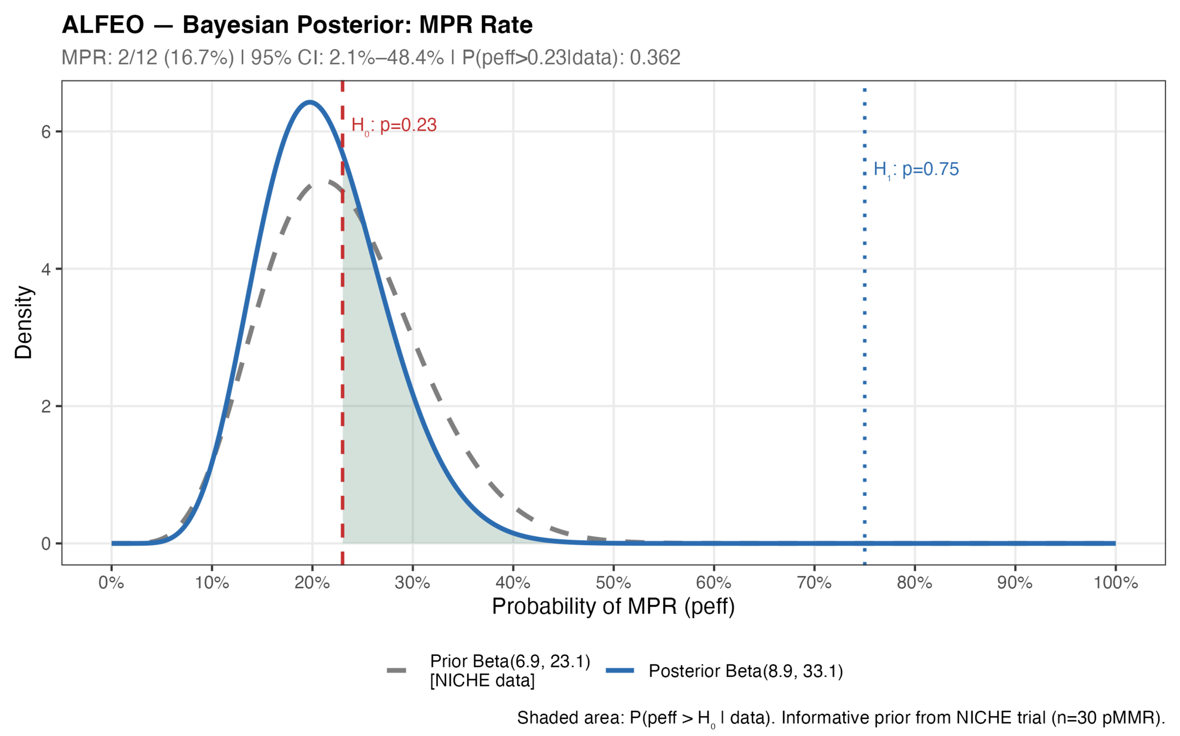
Figure 1 - Clinical results: observed MPR (A, B) and posterior probability of MPR after ICI+HDVC



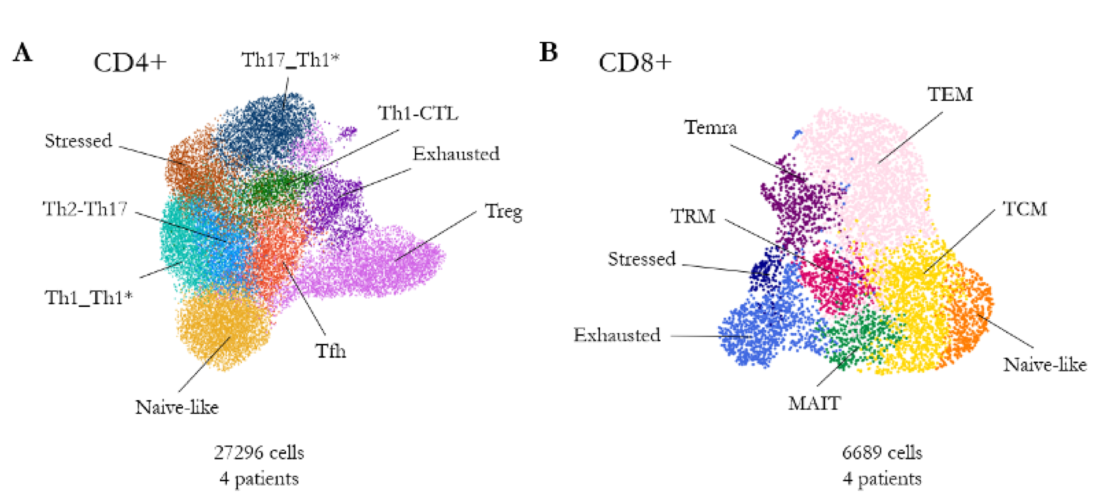
A: Tumor complete regression in MSS responder (A01-06)



B: Lymphnode complete regression in the MSS responder (A01-06)



C: Posterior Probability of MPR conditioned on prior Niche data and observed MPR in trial



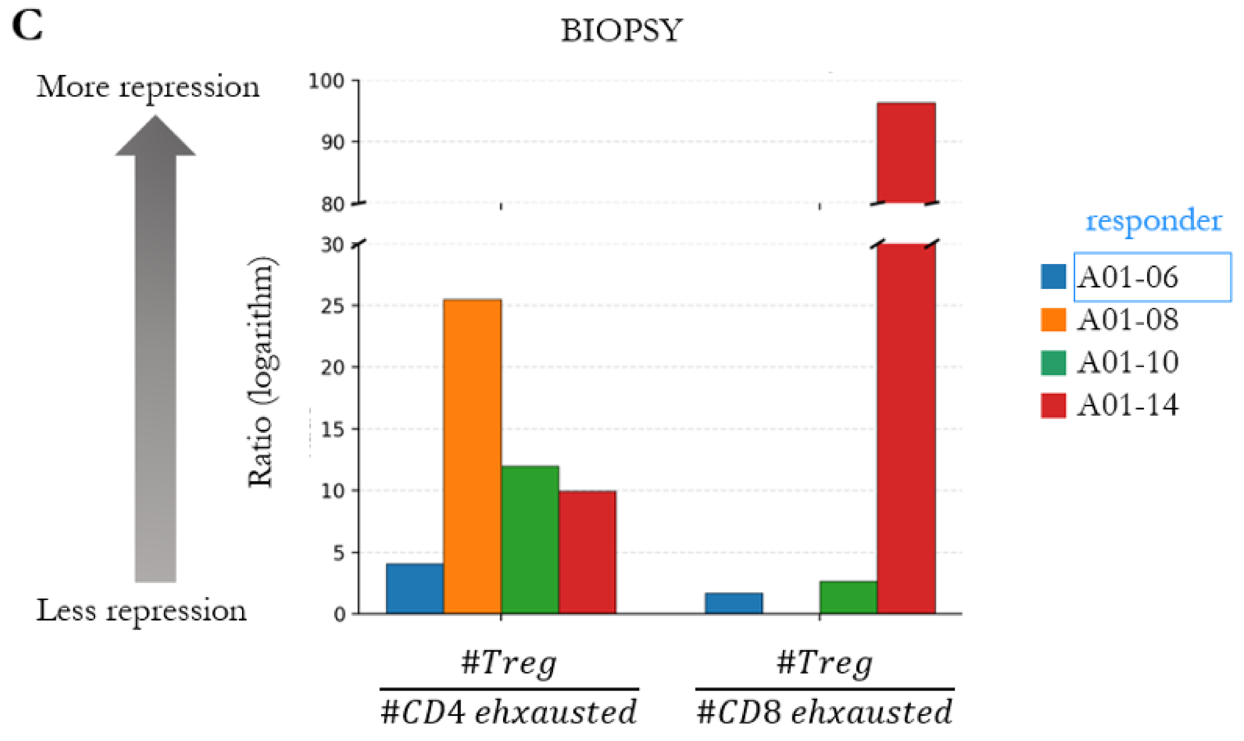


Figure 2. UMAP of the 27,296 CD4+ T and 6,689 CD8+ T sequenced from the 4 patients with complete set (panels A-B), with annotation of the 9 CD4+ clusters and 7 CD8+ clusters (TRM, exhausted, Treg, naïve, central memory, effector memory). Panel C: bar/box plot of Treg/exhausted CD4+ and CD8+ ratio in pre-ICB biopsies, stratified by outcome (1R vs 3NR); The responder patient shows the lowest ratio.

TRANSLATIONAL RESULTS

scRNA/TCR-seq was done on 4 patients with complete paired sets (1 responder [pathological CR], 3 non-responders): 27,296 CD4+ and 6,689 CD8+ T cells; 9 CD4 and 7 CD8 clusters.

The responder showed in the pre-treatment biopsy:

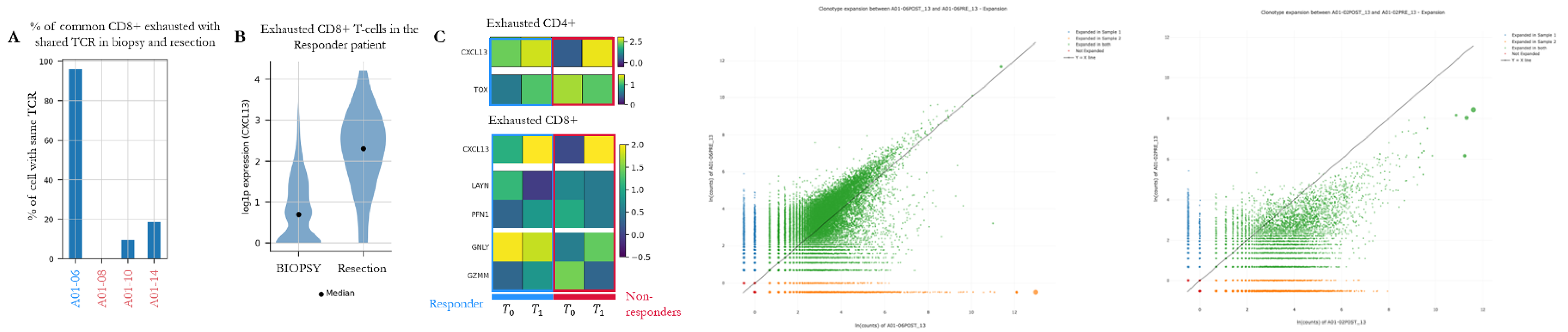
- **Lowest Treg / exhausted-T ratio** and **highest activated exhausted CD4+/CD8+ abundance** pre-ICI.
- **Clonally expanded CD8+ tissue-resident memory (TRM) and exhausted CXCL13+/LAYN+/GNLY+ cells were already present before treatment**, increasing post-ICI; absent in non-responders (TOX-high/GZMM/PFN1 phenotype).
- **Data confirmed tissue (not peripheral) origin** of reactive clones: tumour CD8+ exhausted clones undetectable in pre-ICI PBMC. We observed a near-complete remodelling of the persistent peripheral repertoire (Morisita-Horn **0.097** vs MSS-NR mean **0.699**; z=−2.45), with a post-treatment dominant private clone and low-frequency clones **VDJdb-matched to tumour antigens** (KRAS-mut, SF3B1, gp100, MLANA).

Resistance signatureswere observed in non-responders:

- Persistent highly suppressive, tissue-resident **Treg (ENTPD1+/CTLA4+/CCR8+)** pre- and post-ICB.
- Tumour-exclusive **HLA-G** expression (non-classical MHC-I) — candidate immune-evasion mechanism.

Tertiary lymphoid structures (AI pathology): post-treatment enrichment of **mature TLS** in non-relapsing patients vs absence in the relapsed case, paralleled by increased TILs in non-relapsers.

Figure 3 - Left - A: Exhausted CD8+ clonal persistence between pre-ICB biopsy and post-ICB resection in responders vs non-responders. B: violin/feature plot of CXCL13 expression in exhausted CD8+ tumor clusters, stratified by outcome and timepoint (T0 biopsy, T1 resection). Panel C: heatmap of differential markers between responders vs. non-responders in exhausted CD8+ (CXCL13, LAYN, GNLY vs TOX, GZMM, PFN1) and exhausted CD4+. Right: pre vs post-treatment scatter of TCR-β clonic frequencies in the MSS responder (A01-06) vs mean of non-responders.



CONCLUSIONS

- Adding HDVC to nivolumab + ipilimumab did **not** increase MPR in pMMR/MSS CC.
- Response is governed by a **pre-existing, reactivatable tissue-resident CD8+ CXCL13+/LAYN+ exhausted program**, while resistance maps to **CCR8+/ENTPD1+ Tregs and tumour HLA-G**.
- These data identify **pre-treatment microenvironment biomarkers** (CD8+ CXCL13+/LAYN+ TRM, Treg phenotype, HLA-G, mature TLS) to select MSS/pMMR patients for ICI-based combinations.
- Banked patient-derived organoids and ongoing tissue/ctDNA WES will support future biomarker-driven and co-clinical studies.

References

1. Chalabi M, et al. Neoadjuvant immunotherapy in mismatch-repair–proficient and –deficient colon cancer (NICHE). *Nat Med* 2020;26:566–576.
2. Chalabi M, et al. Neoadjuvant nivolumab + ipilimumab in MSS colon cancer (NICHE-2/3). *Nature* 2025.
3. Magrì A, et al. High-dose vitamin C enhances cancer immunotherapy. *Sci Transl Med* 2020;12:eaay8707.
4. Amatu A, et al. First safety analysis of the ALFEO pilot study. *Ann Oncol* 2024;35(suppl_2):S51–S52.

ACKNOWLEDGEMENTS

EudraCT n. 2022-502101-15-00. Funded by EU NextGenerationEU, "Valorizzazione e potenziamento della Ricerca Biomedica nel SSN", M6, C2, I2.1; CUP H43C21000130006, and Fondazione Oncologia Niguarda ETS

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Ministero della Salute | Finanziato dall'Unione europea NextGenerationEU
PNRR: M6/C2_CALL 2022 Full Proposal
Project Code: PNRR-MAD-2022-12376593
Call section: Malattie Croniche non Trasmissibili (MCnT) ad alto impatto sui sistemi sanitari e
Applicant Institution: Regione Lombardia - Direzione Generale Sanità
Applicant/PI Coordinator: Sartore Bianchi Andrea

Sistema Socio Sanitario
Ospedale Niguarda Cancer Center
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